

DAY 1 Practice Exercises

Exercise: 1 Consider the function $f(x, y) = \sqrt{x+1} - \log_2(4 - x^2 - y^2)$.

- (a) Evaluate $f(0, 0)$, $f(0, \sqrt{2})$, $f(3, -1)$ and $f(1, -1)$
- (b) Find the Domain of the function f , and **sketch it**.

Exercise: 2A Consider the function $f(x, y, z) = \sqrt{9 - x^2 - y^2 - z^2}$

- (a) Evaluate $f(0, 0, 0)$, $f(0, \sqrt{2}, 2)$ and $f(1, 3, 0)$
- (b) Find the Domain of the function f and sketch it.

Exercise: 2B Find the Domain of the function f and sketch it.

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2 - 4}$$

Exercise: 2C Find the Domain of the function f and sketch it.

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2 - 4}}$$

Exercise: 2D Find the Domain of the function f and sketch it.

$$f(x, y, z) = \frac{2x + y^2 - 1}{\sqrt{4 - (x^2 + y^2 + z^2)}}$$

Exercise: 3 Consider the function $f(x, y) = \sin\left(\frac{x+y}{\sqrt{x^2-y}}\right)$

[note that if $z = \frac{x+y}{\sqrt{x^2-y}}$ is real, then it will be in the domain of the *sine* function]

(a) Evaluate $f(0, 0)$, $f(\sqrt{2}, 0)$ and $f(1, -1)$

(b) Find the Domain of the function f .

Exercise: 4A Find the Domain of the function f and sketch it.

$$f(x, y) = \sin\left(\frac{x+y}{\sqrt[9]{x^2-y}}\right)$$

Exercise: 4B Find the Domain of the function f and sketch it.

$$f(x, y) = \sin\left(\sqrt[9]{x^2-y}\right)$$

Exercise: 5a Consider the function $f(x, y) = \sqrt[3]{x+1} - e^{\sqrt{2-y}} + \ln \sqrt{x^2 + y^2 - 9}$.

Find the Domain of the function f and sketch it.

Exercise: 5b Consider the function $f(x, y) = \sqrt[4]{2-x} - e^{\sqrt[3]{2-y}} + \ln \sqrt{x^2 + y^2 - 9}$.

Find the domain and sketch it.

Exercise: 6 Consider the function $f(x, y) = \sqrt[3]{x+1} - \sqrt[4]{1-y} + \ln \sqrt{x^2 + y^2 - 4}$.

Find the Domain of the function f and sketch it.